

MTT-I
SET-A

Academic Year 2024-25

(Odd Semester)

Course	:	B.Tech.	Date	:	06 Nov.2024
Semester/ Section	:	I (Common to all)	Time Duration	:	1:30 Hrs.
Subject & Subject Code	:	Engineering Mathematics-I & 1FY2-01	Max. Marks	:	50

Course Outcomes	
CO1	Recognize fundamental concepts of improper integrals, Beta and Gamma functions and their properties. Application of multiple integrals in finding the areas, volume enclosed by several curves.
CO3	Identify odd, even and periodic function to express them in Fourier series and its application to find sum of infinite series.

Q. No.	CO	Questions	Marks
PART- A: Attempt All Questions (10x2 = 20 Marks)			
1.	CO-1	Evaluate $\int_0^{\pi} \sin^6 x dx$	2
2.	CO-1	Evaluate $\int_1^a \int_1^b \frac{dx dy}{xy}$	2
3.	CO-1	Evaluate $\int_0^a \int_0^b (x^2 + y^2) dx dy$.	2
4.	CO-1	Prove that $\Gamma(n+1) = n\Gamma(n)$	2
5.	CO-1	Find the area of the surface generated by revolving the curve $x = y^3$ about the y-axis from $y = 0$ to $y = 2$.	2
6.	CO-3	Find a_0 for the function $f(x) = \sqrt{1 - \cos x}; -\pi < x < \pi$	2
7.	CO-3	Find a_n for the function $f(x) = ix; -l < x < l$	2
8.	CO-3	Find a_0 for the function $f(x) = \frac{1}{4}(\pi - x)^2, 0 < x < 2\pi$	2
9.	CO-3	Obtain the sine series for unity in $(0, \pi)$	2
10.	CO-3	Write Euler's formula for a Fourier series of a function $f(x)$.	2

PART-B: Descriptive /Analytical / Problem Solving/ Design questions
Attempt Any Three Questions (3x5=15 Marks)

1.	CO-1	Prove that $\int_0^1 \sqrt{1-x^4} dx = \frac{(\Gamma(\frac{1}{4}))^2}{6\sqrt{2}\pi}$	5
2.	CO-1	Evaluate $\int \int xy dx dy$ over the region in the positive quadrant for which $x+y \leq 1$.	5
3.	CO-1	Find the surface of the solid generated by revolution of the ellipse $x^2 + 4y^2 = 16$ about its major axis.	5
4.	CO-1	Find the volume of the solid generated by revolution of the astroid $x^{\frac{2}{3}} + y^{\frac{2}{3}} = a^{\frac{2}{3}}$ about x-axis.	5

PART-C: Descriptive /Analytical / Problem Solving/ Design questions
Attempt Any Three Questions (3x5=15 Marks)

1.	CO-3	Find the Fourier series for the function defined as $f(x) = \begin{cases} -1 & -\pi \leq x < 0 \\ 0 & x = 0 \\ 1 & 0 < x \leq \pi \end{cases}$ $a=0, b=\pi$	5
2.	CO-3	Find b_n for the function $f(x) = x - x^2$ from $x = -1$ to $x = 1$	5
3.	CO-3	Find the Fourier series for the function defined as $f(x) = \begin{cases} x & \text{for } 0 \leq x < \pi \\ 2\pi - x & \text{for } \pi \leq x < 2\pi \end{cases}$	5
4.	CO-3	Find the half range cosine series for the following function $f(x) = (x-1)^2, 0 < x < 1$	5

$$\int_0^{\infty} x^n e^{-ax} dx = \frac{n!}{a^{n+1}}$$

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